Kiarad Zafar Heidari

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EDUCATION

University of Waterloo Waterloo, ON

Bachelor of Science in Honours Mathematical Physics *Sep. 2025 - Present*

TECHNICAL SKILLS

Languages: Python, C, C++, Go, TypeScript, SQL

Frameworks & Tools: PyTorch, TensorFlow, Keras, OpenCV, scikit-learn, NumPy, Pandas, LangGraph, React, Next.js, Node.js, Git, Docker, Linux

Embedded: ESP32, ESP-IDF, Arduino, PlatformIO, NRF24, Microcontroller Firmware, Embedded Networking

EXPERIENCE

Machine Learning Research Assistant May 2026 - Present

Center for Advanced Diffusion-Wave Technologies, University of Toronto *Toronto, ON*

- Engineered a **vectorized exact-ODE solver in PyTorch** for 77,500 thermal-video pixels, cutting full-image fitting from **3.2 hours to under 10 minutes** versus per-pixel PINNs.
- Built a hybrid physics-neural model combining analytical recursion, a shared 305-parameter temporal residual network, and per-pixel correction amplitudes, reducing **full-image MSE by 20%** vs exact-ODE.
- Validated **region-based thermal defect detection** on a known steel defect, achieving **0.13 mm centroid error and 0.82 IoU**; diagnosed a flat-field preprocessing artifact and corrected the detection protocol.

Lead Manager & Creative Director 2021 - 2023

Hazard Esports *Remote*

- Directed operations across NA, EU, and Caribbean teams, growing the organisation to **10,000+ subscribers** and negotiating sponsorships that generated **$2,500+ in revenue** through community campaigns.

PROJECTS

AgentHarness | *Python, TypeScript, FastAPI, PostgreSQL, pgvector, RAG, Docker* | GitHub

- Built a **self-hosted ReAct runtime** with **multi-agent delegation**, a streaming TypeScript terminal UI, and a FastAPI interface.
- Implemented **persistent memory and hybrid RAG** using PostgreSQL/pgvector, full-text/vector retrieval, and byte-exact archival of evicted context.
- Prototyped **learned memory management** with a **6-action offline RL trainer**; added persona-conditioned retrieval and Soul Spec adapters for agent portability.

MLscratch -- Neural Network Library from Scratch | *C, Linear Algebra* | GitHub

- Built a three-layer MLP and **topological-sort autograd engine in pure C**, achieving **~98% MNIST accuracy** with 13,002 parameters on 60k training samples.
- Implemented a **cross-platform arena allocator** using Linux mmap/mprotect and Windows VirtualAlloc, reserving 1 GiB of address space with on-demand commitment; added a PCG-based PRNG seeded from OS entropy.

Siamese Facial Verification | *Python, TensorFlow, Keras, OpenCV* | GitHub

- Built a **Siamese CNN for facial verification** with shared-weight convolutional branches, 4,096-dimensional embeddings, and a custom L1-distance layer to predict match probabilities.
- Trained on **6,000 image pairs** over 50 epochs using Adam and binary cross-entropy; deployed a **live webcam app with dual thresholds** to tune false-accept and false-reject trade-offs without retraining.